

MATH 215 SI

Session 1

Session 1

Lectures 1 - 4

- ▶ 12.1 Three-Dimensional Coordinate Systems
- ▶ 12.2 Vectors
- ▶ 12.3 The Dot Product
- ▶ 12.4 The Cross Product
- ▶ 12.5 Equations of Lines and Planes

12.1 Three-Dimensional Coordinate Systems

- ▶ Defined as $\mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$
- ▶ We represent any point of space by an ordered triple (a, b, c)

Exercises

12.1 Three-Dimensional Coordinate Systems

- ▶ What does the equation $x = 4$ represent in $\mathbb{R}^2$?
- ▶ What does the same equation represent in $\mathbb{R}^3$?
- ▶ Describe and/or sketch the surface in $\mathbb{R}^3$ represented by the equation $x + z = 2$

12.2 Vectors

We will define a vector $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$

- ▶ Then the magnitude of $\mathbf{v}$ is given by $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$
- ▶ Then the unit vector (direction) of $\mathbf{v}$ is given by $\hat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$
- ▶ Vector addition is given by $\mathbf{v} + \mathbf{u} = \langle v_1 + u_1, v_2 + u_2, \dots, v_n + u_n \rangle$
- ▶ Scalar multiplication by c is given by $c\mathbf{v} = \langle cv_1, cv_2, \dots, cv_n \rangle$
- ▶ The standard basis vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are unit vectors that point in the direction of the positive x -, y -, and z -axes.
- ▶ Recall the Parallelogram Law

12.2 Vectors

cont.

Properties of Vectors

- ▶ $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$
- ▶ $\mathbf{a} + \mathbf{0} = \mathbf{a}$
- ▶ $c(\mathbf{a} + \mathbf{b}) = c\mathbf{a} + c\mathbf{b}$
- ▶ $(cd)\mathbf{a} = c(d\mathbf{a})$
- ▶ $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$
- ▶ $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$
- ▶ $(c + d)\mathbf{a} = c\mathbf{a} + d\mathbf{a}$
- ▶ $1\mathbf{a} = \mathbf{a}$

Exercises

12.2 Vectors

- ▶ Generate an expression that is equivalent to the vector $\langle 11, 3, 5 \rangle$ as a linear combination of $\mathbf{v} = \langle 5, 1, 2 \rangle$ and the standard basis vectors (each vector should appear in your expression).
- ▶ Find a vector that has the same direction as $\langle 6, 2, -3 \rangle$ but has length 4.

12.3 The Dot Product

If $\mathbf{a} = \langle a_1, a_2, \dots, a_n \rangle$ and $\mathbf{b} = \langle b_1, b_2, \dots, b_n \rangle$, then the dot product of $\mathbf{a}$ and $\mathbf{b}$ is the number $\mathbf{a} \cdot \mathbf{b}$ given by $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$

- ▶ The dot product measures how aligned two vectors are
- ▶ If θ is the angle between vectors $\mathbf{a}$ and $\mathbf{b}$, then $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$
- ▶ Two vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal $\iff \mathbf{a} \cdot \mathbf{b} = 0$

12.3 The Dot Product

cont.

Properties of the Dot Product

- ▶ $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$
- ▶ $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- ▶ $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
- ▶ $(c\mathbf{a}) \cdot \mathbf{b} = c(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (c\mathbf{b})$
- ▶ $\mathbf{0} \cdot \mathbf{a} = 0$

12.3 The Dot Product

cont.

Projections

- ▶ The scalar projection of $\mathbf{b}$ onto $\mathbf{a}$ is given by $\text{comp}_a \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|}$
- ▶ The vector projection of $\mathbf{b}$ onto $\mathbf{a}$ is given by $\text{proj}_a \mathbf{b} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|} \right) \frac{\mathbf{a}}{|\mathbf{a}|}$

Exercises

12.3 The Dot Product

- ▶ Let $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$, and let θ_1 be the angle $\mathbf{v}$ makes with the x -axis, θ_2 be the angle $\mathbf{v}$ makes with the y -axis, and θ_3 be the angle $\mathbf{v}$ makes with the z -axis. Evaluate the following expression: $\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3$

12.4 The Cross Product

If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ then the cross product of $\mathbf{a}$ and $\mathbf{b}$ is the vector $\mathbf{a} \times \mathbf{b}$ given by

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

- ▶ $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$
- ▶ $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta = \text{area of the parallelogram determined by } \mathbf{a} \text{ and } \mathbf{b}$
- ▶ Two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel $\iff \mathbf{a} \times \mathbf{b} = \mathbf{0}$

12.4 The Cross Product

cont.

Properties of the Cross Product

- ▶ $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
- ▶ $(c\mathbf{a}) \times \mathbf{b} = c(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (c\mathbf{b})$
- ▶ $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
- ▶ $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$
- ▶ $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

Exercises

12.4 The Cross Product

- Find the cross product $\mathbf{a} \times \mathbf{b}$ and verify that it is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$. Let $\mathbf{a} = 2\mathbf{j} - 4\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 3\mathbf{j} + \mathbf{k}$

12.5 Equations of Lines and Planes

A line through point $\mathbf{r}_0$ with direction $\mathbf{v}$ is given by $\mathbf{r}(t) = t\mathbf{v} + \mathbf{r}_0$

- ▶ Note that $\mathbf{r}$ is a function of t
- ▶ By eliminating t we can yield the symmetric equations (not useful in my experience).

A plane through point $\mathbf{r}_0$ with normal vector $\mathbf{n}$ is given by $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$

- ▶ Note that $\mathbf{r}$ is an arbitrary vector
- ▶ Expanding the expression we yield the scalar equation of the plane given by $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

12.5 Equations of Lines and Planes

cont.

Distances

- The distance D from point $P_1(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is given by $D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$

Exercises

12.5 Equations of Lines and Planes

Suppose P_1 is a plane that contains the points $(0, 3, 0)$, $(-3, 0, 0)$, and $(2, 6, 1)$.

- Find the equation of the plane P_1

Let P_2 be the plane that contains the point $(0, 2, 1)$ and the line $\mathbf{l}(t) = \langle 2t, t, 1 + 3t \rangle$

- Find the angle between P_1 and P_2

Fin

This concludes session 1.